TEXORA · LECTURE SHEET

Fabric Buckling, Shear and Drape

The three low-stress deformations that decide how a cloth handles, tailors and hangs: bending and its instability buckling, in-plane shear on the yarn trellis, and drape — the visible sum of the other two, measured by Cusick's coefficient.

FM-4203 · FABRIC MECHANICS · TOPIC 4 · FABRIC MANUFACTURING SPECIALIZATION

1. Bending: bending length and flexural rigidity

A fabric bends far more easily than it stretches, because bending asks yarns only to curve and crossovers to rotate slightly. The standard low-stress bending measurement is the **cantilever test** (Shirley stiffness tester): a strip slides over an edge until its tip droops to a fixed angle (41.5°), and the overhang length L at that moment gives the **bending length** c through

$$c = (L \div 2) \times [\cos(\theta \div 2) \div \tan\theta]^{1/3} \quad \rightarrow \quad \text{at } \theta = 41.5°,\ c = L \div 2$$

The angle 41.5° is chosen precisely so that the bracket equals 1 — a small elegance of the standard method. The bending length combines stiffness and weight: a heavy limp cloth and a light stiff one can share a c. To isolate the material's stiffness, use the **flexural rigidity**:

$$G = 0.1 \times M \times c^3 \quad (\mu N \cdot m), \quad M = \text{fabric mass in } g/m^2,\ c \text{ in cm}$$

WORKED EXAMPLE — CANTILEVER BENDING

A shirting strip reaches the 41.5° line at an overhang L = 7.4 cm, so its bending length is c = 7.4 ÷ 2 = **3.7 cm**.

A suiting of M = 200 g/m² with c = 4.0 cm has flexural rigidity G = 0.1 × 200 × 4.0^3 = 0.1 × 200 × 64 = **1280 µN·m ≈ 1.28 mN·m**.

Note the cube: a 25% rise in bending length nearly doubles G ($1.25^3 \approx 1.95$). Small construction changes — a slightly closer sett, a stiffer finish — are levered into large stiffness changes, which is why bending is the most sensitive single handle property, and why warp and weft values are always reported separately: c follows the crimp and cover of the direction being bent.

2. Buckling: bending's instability

Bending becomes **buckling** when a fabric is compressed end-to-end (in its own plane) or bent past the point where the compressed face can shorten: the cloth escapes sideways into waves rather than

compressing. The physics is Euler's column instability scaled down to a sheet — the buckling load rises steeply with flexural rigidity and falls with the unsupported length — but in textiles buckling appears less as collapse than as familiar defects and behaviours:

- **Creasing and wrinkling** are local buckling: the inside of a sharp bend is a short compressed column of cloth, and it waves. Recovery from creasing is the reverse buckling — driven by the elastic energy stored in the bent yarns — which is why crease recovery (TE-3201) tracks bending elasticity and resin crosslinking.
- **Seam pucker** is buckling of the fabric strip alongside a seam, where the sewing thread's tension compresses the cloth in-plane.
- **Bagging** at knees and elbows is buckling plus incomplete recovery under repeated biaxial strain.
- **Tailoring:** the ease with which a cloth can be moulded (a sleeve head eased into an armhole) is controlled compression — the tailor is deliberately working just below the buckling threshold.

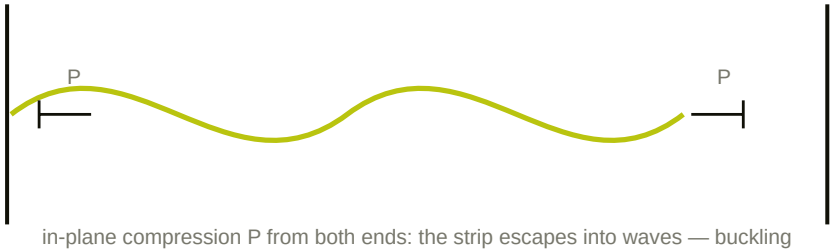


Figure 1 — Buckling of a fabric strip under end compression. The wavelength and the load at which waves appear both rise with flexural rigidity; a limp cloth buckles into many fine wrinkles, a stiff one into few broad folds — the visible difference between creasing in a lining and in a canvas.

3. Shear: the trellis mechanism

Pull a woven cloth on the bias, or push one edge parallel to the other, and the warp and weft **rotate at their crossovers like the bars of a garden trellis**: the yarns themselves barely stretch, the angles between them change. This is fabric **shear**, measured by the **shear angle** γ — the change in the originally right angle between warp and weft. Shear is opposed first by friction at crossovers (low rigidity, with hysteresis), then, as the yarns swing round and pack against each other side by side, by jamming: the **locking angle**, beyond which the cloth can shear no further without buckling out of plane.

Shear is what makes a flat cloth able to cover a doubly curved body at all: a woven fabric conforms to a shoulder or a car seat not by stretching but by trellising — yarns rotating toward the direction of pull — until the locking angle is reached, whereupon it wrinkles. Formability in composites moulding and in garment making is, quantitatively, the available shear range before locking.

> **WORKED EXAMPLE — SHEAR ANGLE FROM A BIAS-EXTENSION TEST**

A strip cut at 45° (bias) to warp and weft, gauge length 20.0 cm, extends under small load to 22.4 cm: extension ratio $\lambda = 22.4 \div 20.0 = 1.12$.

For the trellis, the strip length follows the diagonal of the rotating square: $\lambda = \sqrt{2} \times \cos(45° - \gamma \div 2)$, so $\cos(45° - \gamma \div 2) = 1.12 \div 1.414 = 0.792$, giving $45° - \gamma \div 2 = 37.6°$ and $\boldsymbol{\gamma \approx 14.7°}$.

A 12% bias extension has cost a shear of about 15° — cheap deformation, exactly as the mechanism promises. The same specimen pulled on-grain would extend ~1–2% at the same load, all of it crimp interchange (Topic 3) and yarn strain.

4. Drape: bending and shear acting together

Drape is the deformation of a cloth under its own weight when supported over only part of its area — the way a skirt hangs, a tablecloth falls, a curtain folds. It is not a new mechanism: the hanging parts bend (Section 1) and the transition zones shear (Section 3). The standard measure is **Cusick's drape coefficient** from the drapemeter: a circular specimen (30 cm diameter) is laid on a smaller support disc (18 cm diameter) and the area of its projected shadow A_s is compared with the specimen area A_p and the disc area A_d:

$$F = (A_s - A_d) \div (A_p - A_d) \times 100\%$$

WORKED EXAMPLE — DRAPE COEFFICIENT

Specimen radius 15 cm: $A_p = \pi \times 15^2 = \mathbf{706.9\ cm^2}$. Disc radius 9 cm: $A_d = \pi \times 9^2 = \mathbf{254.5\ cm^2}$.

The shadow of the draped specimen measures $A_s = 420\ cm^2$.

$F = (420 - 254.5) \div (706.9 - 254.5) \times 100 = 165.5 \div 452.4 \times 100 \approx \mathbf{36.6\%}$.

Reading the scale: $F \to 100\%$ is a board (the shadow is the whole specimen), $F \to 0\%$ is a fluid cloth collapsing onto the disc. A value near 37% is a moderately draping fabric; the accompanying **node count** (the number of folds around the disc) completes the description — few deep nodes for a stiff cloth, many shallow ones for a limp cloth of the same F.

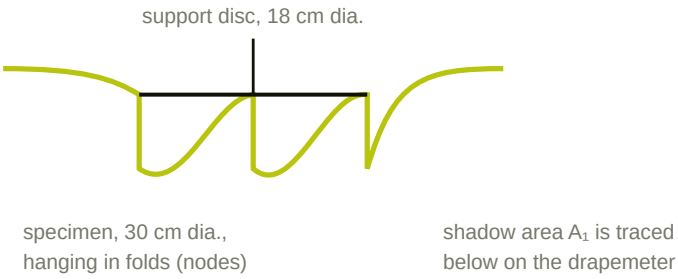


Figure 2 — The Cusick drape test in section: the annulus of specimen beyond the support disc falls into folds whose number and depth are set by bending rigidity (fold depth) and shear rigidity (how the folds distribute). The projected shadow area gives F.

The three properties of this sheet, with tensile behaviour, form the **low-stress mechanical fingerprint** of a fabric — the set measured by the KES-F and FAST systems met in TE-3201 — because they, not the breaking load, are what a hand feels, what a tailor fights, and what a customer sees on the rail.

Glossary

Bending length c — overhang-based stiffness index; at the standard 41.5°, c = L/2. **Flexural rigidity G** — 0.1 × M × c³ μN·m; the weight-free bending stiffness. **Buckling** — sideways escape of a cloth under in-plane compression; root of creasing, pucker and bagging. **Shear angle γ** — change in the warp–weft angle under trellis rotation. **Locking angle** — shear angle at which yarns jam side by side. **Drape coefficient F** — Cusick's shadow-area ratio, 0% fluid to 100% rigid.

Key points

• Cantilever: c = L/2 at 41.5° (worked 3.7 cm); G = 0.1Mc³ — the cube makes rigidity the most construction-sensitive handle property (worked 1280 μN·m).
• Buckling is bending under compression: creasing, seam pucker, bagging and tailor's moulding are one instability at different scales.
• Shear is trellis rotation of nearly inextensible yarns: a 12% bias extension ≈ 15° of shear; formability ends at the locking angle.
• Drape = bending + shear under gravity, measured by Cusick's F (worked 36.6%) with the node count as its necessary companion.

Practice

1. A fabric of mass 150 g/m² has bending length 3.0 cm in the warp direction. Compute its flexural rigidity, and the new rigidity if a finish raises c to 3.6 cm.
2. In a drape test (same geometry as the worked example) the shadow area is 600 cm². Compute F and characterise the fabric.
3. Explain why a bias-cut panel conforms to a curved seam without puckering, while a straight-grain panel puckers, using the trellis mechanism and the locking angle.
4. A tailor complains that a cloth "creases if you look at it". Connect the complaint to two measurable properties from this sheet and one from fibre level (Topic 1).

Answer key

1. G = 0.1 × 150 × 3.0³ = 0.1 × 150 × 27 = **405 μN·m**. At c = 3.6: G = 0.1 × 150 × 46.7 = **700 μN·m** — a 20% rise in c gives a 73% rise in rigidity (1.2³ = 1.73).

2. F = (600 − 254.5) ÷ (706.9 − 254.5) × 100 = 345.5 ÷ 452.4 × 100 ≈ **76%** — a stiff, poorly draping fabric (canvas/coating territory): the annulus largely holds itself out rather than falling into folds.
3. The bias panel meets the curve by **shear**: warp and weft rotate at crossovers, letting the panel change shape in-plane at almost no stress, so no excess length accumulates to buckle. The straight-grain panel can only comply by crimp interchange and yarn strain — a few percent at most — so the surplus length along the seam compresses the cloth in-plane and it escapes as pucker (buckling). Puckering arrives earlier if the panel's required shear exceeds the fabric's locking angle.
4. (i) **Low flexural rigidity / low bending recovery** — the cloth buckles into creases under trivial compression (Section 2) and stores little elastic energy to spring back. (ii) **Low crease recovery angle**, the direct measure of that reverse buckling. (iii) Fibre level: a fibre with low elastic recovery and high plastic set (e.g. a high-amorphous-content fibre whose hydrogen bonds re-form in the bent position — viscose, or cotton un-crosslinked) locks the buckle in; resin crosslinking or a resilient fibre (wool, polyester) reverses it.
